

SMARTPHONE MARKET INTELLIGENCE ANALYSIS USING SQL

Dataset: smartphones

Rows: 980

Columns: 26

Project Objective

Smartphone Market Intelligence Analysis Using SQL

The objective of this project is to analyze a smartphone market dataset containing approximately 980 smartphone models and generate actionable business insights using SQL. The project focuses on understanding smartphone pricing strategies, brand positioning, customer ratings, feature adoption trends, product segmentation, and competitive market dynamics.

Through a series of 35 business-driven SQL analyses, the project aims to answer key questions such as:

- Which brands dominate different price segments?
- How does pricing relate to ratings and specifications?
- Which brands offer the best value for money?
- How has 5G adoption evolved across brands?
- What factors contribute to premium smartphone positioning?
- Which smartphones provide the strongest overall feature combination?
- How do brands differ in portfolio diversification and pricing consistency?

The project leverages advanced SQL techniques including joins, subqueries, common table expressions (CTEs), window functions, ranking functions, segmentation analysis, and statistical measures to transform raw smartphone data into meaningful business intelligence.

Analyze smartphone pricing strategies, feature adoption trends, brand positioning, processor performance, battery trends, camera competition, and value-for-money offerings using SQL.

SQL Concepts Covered:

- SELECT
- WHERE
- ORDER BY
- GROUP BY
- HAVING
- CASE WHEN
- Aggregate Functions
- Subqueries
- Correlated Subqueries
- CTEs
- Window Functions
- RANK()
- DENSE_RANK()
- ROW_NUMBER()
- PERCENT_RANK()
- NTILE()
- LAG()
- LEAD()
- Conditional Aggregation
- Business KPI Creation
- Market Segmentation

LEVEL 1 – BASIC ANALYSIS

Q1. Total Smartphone Models

How many smartphone models exist in the dataset?

Insight : The dataset contains 980 unique smartphone records with no duplicate brand-model combinations. This indicates good data quality and provides a reliable foundation for subsequent analysis.

Q2. Brand Catalogue Overview

List all smartphone brands and the number of models offered by each brand. Sort by model count descending.

Insights : Xiaomi (134 models) and Samsung (132 models) have the largest smartphone portfolios in the dataset, followed by Vivo, Realme, and Oppo. This suggests that major brands compete across multiple price segments, whereas smaller brands maintain a more focused product range.

Q3. Budget 5G Phones

Find all smartphones priced below ₹15,000 that support 5G. Display brand, model, price, and rating.

Insight: A total of **46 smartphones** in the dataset offer 5G connectivity at a price below ₹15,000, indicating that 5G technology has become increasingly accessible in the budget segment. **Realme (10 models), Xiaomi (9 models), and Poco (8 models)** are the dominant players, collectively accounting for more than half of all budget 5G devices. This suggests that these brands are aggressively targeting price-sensitive consumers by offering affordable 5G smartphones. In contrast, premium-focused brands have a limited presence in this segment, highlighting the strong competition among value-oriented manufacturers.

Q4. Missing Rating Audit

How many smartphones have missing ratings? What percentage of the total catalogue does this represent?

Insight:

The dataset contains 101 smartphones with missing ratings, accounting for 10.31% of all records. Since nearly 90% of smartphones have valid ratings, rating-based analyses remain reliable, although unrated devices should be handled appropriately during analysis.

Q5. Market Benchmark Metrics

Calculate the overall average:

- Price
- Rating
- Battery Capacity
- RAM
- Internal Storage

Insight:

The average smartphone in the dataset costs ₹32,520, has a rating of 78.26, offers 4,818 mAh battery capacity, 6.56 GB RAM, and 141 GB storage. These figures establish the baseline benchmarks for comparing brands and identifying premium or value-for-money smartphones.

Business Interpretation

This question is important because it creates the **market baseline** for the entire project.

For example:

- Any phone priced above **₹32,520** can be considered above-average in price.
- Any phone rated above **78.26** is performing better than the average smartphone.
- Phones with more than **6.56 GB RAM** or **141 GB storage** can be considered better equipped than the average device.

LEVEL 2 – BUSINESS ANALYSIS

Q6. Price Segment Distribution

Classify smartphones into:

- Budget (< ₹15,000)
- Mid-Range (₹15,000–₹40,000)
- Premium (₹40,000–₹80,000)
- Flagship (> ₹80,000)

Show:

- Number of phones
- Average rating

for each segment.

Insight:

The Mid-Range segment dominates the market with 432 smartphones, representing the largest share of the catalogue. Premium devices achieve the highest average rating (84.62), indicating strong customer satisfaction. Budget smartphones receive the lowest ratings, while Flagship devices do not significantly outperform Premium phones despite their higher prices.

Q7. Brand-wise Performance

For each brand, calculate:

- Average Price
- Average Rating

Include only brands with at least 5 smartphones.

Insight: Significant differences exist in brand positioning across the smartphone market. Nothing (85.2), Asus (87.0), and Nubia (84.33) achieve the highest average ratings, indicating strong customer satisfaction, although they offer relatively small product portfolios. Among major brands, OnePlus (82.24) and iQOO (82.22) maintain strong ratings while remaining competitively priced. In contrast, Apple commands the highest average price (₹95,967) but records a lower average rating (76.8) than several Android competitors, suggesting that higher pricing does not necessarily translate into higher customer ratings.

Q8. Premium Brand Identification

Identify brands whose average smartphone price is above the overall market average.

Insights: Sixteen brands have an average smartphone price above the market average of ₹32,520.50. Apple, Samsung, OnePlus, Google, Sony, and Huawei are among the major premium-positioned brands, while Vertu, Royole, and Leitz serve the ultra-premium segment. This highlights the diverse pricing strategies adopted across the smartphone market.

Q9. Value-for-Money Brands

Find brands that provide above-average ratings while charging below-average prices.

Insights: Nine brands qualify as value-for-money brands by achieving above-average ratings while maintaining below-average prices. Doogee (86.0), Nothing (85.2), and BLU (85.0) deliver the highest customer ratings without exceeding the market average price. Among major brands, iQOO, Motorola, Xiaomi, Oppo, and Poco offer strong customer satisfaction at competitive price points, demonstrating an effective balance between affordability and product quality.

Q10. Android vs iOS Comparison

Compare Android and iOS smartphones on:

- Average Price
- Average Rating
- Average Battery Capacity
- Average RAM

Insights: Android dominates the dataset with **909 smartphones (95.2%)**, compared to only **46 iOS devices (4.8%)**. Android phones offer higher average RAM (**6.58 GB vs 5.28 GB**) and larger batteries (**4,874 mAh vs 3,527 mAh**) while costing significantly less on average (**₹27,992 vs ₹95,967**). Despite being over **3.4× more expensive**, iOS devices achieve a slightly lower average rating (**76.80**) than Android smartphones (**78.22**), highlighting Android's strong value proposition in terms of hardware specifications.

Q11. 5G Price Premium

Compare the average price of 5G phones versus non-5G phones.

Insight: The dataset contains 549 5G smartphones and 431 non-5G smartphones, indicating that 5G devices now represent the majority of the smartphone market. On average, 5G smartphones cost ₹43,200, while non-5G smartphones cost ₹18,917, resulting in a price difference of approximately ₹24,284. This means 5G-enabled devices are priced at more than 2.2 times higher than non-5G smartphones, highlighting a significant technology premium associated with 5G connectivity.

With over **56% of smartphones (549 out of 980)** supporting 5G, the market is clearly transitioning toward next-generation connectivity. However, the large price gap indicates an opportunity for brands to expand affordable 5G offerings and capture price-sensitive consumers.

Q12. 5G Adoption by Brand

For each brand, calculate:

- Total Models
- 5G Models
- 5G Adoption Percentage

Insight: Significant differences exist in 5G adoption across smartphone brands. iQOO leads the market with a 93.8% adoption rate (30 out of 32 models), followed by OnePlus (85.7%) and Apple (80.4%), demonstrating a strong focus on next-generation connectivity. Among major-volume brands, Poco (63.4%), Motorola (61.5%), and Xiaomi (56.7%) show relatively high 5G penetration, while Samsung (52.3%), Realme (52.6%), and Vivo (47.8%) maintain a more balanced mix of 5G and non-5G devices. In contrast, Tecno (15.2%), Infinix (24.1%), and Huawei (25.0%) exhibit the lowest adoption rates, indicating a continued focus on budget and legacy device segments.

Q13. Processor Market Share

Which processor brands power the most smartphones?

Insight : Snapdragon dominates the smartphone chipset market with 413 devices (43.0% market share), making it the most widely used processor brand in the dataset. Helio (20.9%) and Dimensity (18.4%) rank second and third, meaning the top three processor families power over 82% of all smartphones. In contrast, Exynos (5.2%) and Apple Bionic (4.7%) hold relatively small shares, while Kirin (0.7%) and Google Tensor (0.9%) remain niche players. The results highlight Qualcomm's market leadership and MediaTek's strong presence in the smartphone chipset ecosystem.

Q14. Fast Charging Brand Analysis

-- Brands with 3+ fast-charging models (non-NUL wattage).

Insight: Nubia leads the fast-charging segment with an average charging speed of 95W, followed by OnePlus (76.3W) and iQOO (75.3W). Xiaomi maintains the largest fast-charging portfolio with 121 models and supports charging speeds up to 210W, while Realme records the highest charging speed in the dataset at 240W. In contrast, Samsung (24.1W) and Apple (20.3W) offer significantly lower charging speeds despite their strong market presence. These findings suggest that Chinese smartphone

brands are driving innovation in charging technology and increasingly use ultra-fast charging as a key product differentiator.

Q15. Camera Comparison by Brand

-- Top 10 brands by average rear camera MP.

Insight : Doogee leads the dataset with the highest average rear camera resolution of 108 MP and an average front camera resolution of 32 MP, although its ranking is based on only 2 smartphone models. Among brands with larger portfolios, Nokia (79.8 MP), Motorola (67.2 MP across 52 models), Honor (66.7 MP), and Xiaomi (65.3 MP across 134 models) demonstrate strong camera specifications. Notably, Xiaomi combines high camera resolution with the largest smartphone portfolio in the dataset, indicating a broad focus on camera-centric devices. These findings suggest that while some niche brands achieve the highest camera specifications, major brands such as Xiaomi and Motorola deliver competitive camera capabilities at scale.

Q16. Refresh Rate vs Price Analysis

-- Custom buckets: 60 / 90 / 120 / 144Hz+

Insight : Smartphones with higher refresh rates command significantly higher prices and ratings. Devices with 120Hz displays average ₹45,996 and a rating of 83.15, while 144Hz+ models average ₹48,128 and achieve the highest rating of 85.4. In comparison, traditional 60Hz phones average ₹25,504 with a rating of 72.98. Notably, 120Hz has become mainstream with 344 devices, nearly matching the 368 phones in the 60Hz segment. These findings suggest that high-refresh-rate displays have evolved from a premium gaming feature into a key differentiator in the modern smartphone market.

Q17. Expandable Storage Impact

-- Does extended memory support mean less built-in storage?

Insight : Smartphones with expandable storage support account for the majority of the dataset (618 devices, 63.1%) but offer significantly lower built-in storage, averaging 108.4 GB. In contrast, phones without expandable storage provide nearly 1.8× more internal storage (196.7 GB) and are priced substantially higher at ₹55,681, compared to ₹18,954 for expandable-storage devices. This suggests that premium smartphones increasingly rely on larger built-in storage capacities rather than memory card expansion, while budget and mid-range devices continue to use expandable storage as a cost-effective way to provide flexibility to users.

LEVEL 3 – INTERMEDIATE ANALYSIS

Q18. Hidden Gem Smartphones

-- Below market avg price AND above market avg rating.

Insight : The analysis identified a large number of highly rated smartphones priced below the market average of ₹32,520, highlighting strong value-for-money opportunities across multiple brands. Xiaomi, Samsung, Oppo, Vivo, Motorola, Realme, and Poco appear most frequently in the hidden-gem segment, demonstrating their ability to deliver premium user experiences at accessible price points. The highest-rated model is the Xiaomi Redmi Note 12 Explorer (Rating: 89, Price: ₹24,999), followed by devices such as the Samsung Galaxy S20 FE 5G (Rating: 88, Price: ₹31,239) and Motorola Edge 20 Fusion 5G (Rating: 87, Price: ₹18,999). These findings suggest that superior customer satisfaction is not limited to flagship devices and can often be found in competitively priced mid-range smartphones.

Q19. Potentially Overpriced Smartphones

-- Above market avg price AND below market avg rating.

Insight : The analysis identified several smartphones that are priced above the market average of ₹32,520 but receive below-average ratings (<78.26). Apple dominates this segment, with multiple premium devices such as the iPhone 14 Pro Max (1TB) priced at ₹182,999 with a rating of 78 and the

iPhone 15 priced at ₹82,990 with a rating of 72. The most extreme case is the Vertu Signature Touch, which costs ₹650,000 but has a rating of only 62. Other brands appearing in this segment include Samsung, Google, Vivo, OnePlus, and Nokia. These findings suggest that higher pricing does not always translate into higher customer satisfaction, highlighting the importance of evaluating value alongside premium positioning.

One analyst note for yourself

For brands like **Vertu**, the high price is driven by luxury materials, exclusivity, and branding rather than smartphone specifications. Therefore, the strongest business takeaway is actually:

Premium pricing alone does not guarantee higher user ratings.

Q20. Battery Capacity Tiers and Ratings

-- Four tiers: Low / Standard / High / Ultra

Insight : Smartphone ratings generally improve as battery capacity increases, up to a certain point. Devices in the High (4500–4999 mAh) segment achieve the highest average rating of 84.38 across 173 models, outperforming all other battery tiers. In comparison, Low battery phones (<3000 mAh) receive the lowest average rating (68.6) across just 13 devices. Although the Ultra (5000+ mAh) segment dominates the market with 600 smartphones (61.2% of the dataset), its average rating (76.5) is lower than the High tier. This suggests that consumers value a balanced combination of battery life, design, weight, and overall performance rather than simply the largest battery capacity.

Q21. NULL Handling in Fast Charging — Data Quality Check

-- Phones that claim fast charging available but wattage is NULL.

Insight: The dataset contains 837 smartphones that claim to support fast charging, of which 769 devices (91.9%) have a valid charging wattage recorded. However, 68 smartphones (8.1%) indicate fast-charging support but have missing wattage information, representing a potential data quality issue. Additionally, all 143 devices without fast-charging support correctly have no wattage value recorded. Overall, the dataset demonstrates strong data completeness, with only a small proportion of fast-charging devices requiring further validation or specification enrichment.

Q22. Processor Speed vs Price Bands

-- Avg speed, price, rating per processor brand. No NULLs.

Insight: Apple's Bionic processors deliver the highest average clock speed at 3.10 GHz, followed by Google Tensor (2.82 GHz) and Huawei Kirin (2.77 GHz). These processor families are primarily found in premium smartphones, with average prices ranging from ₹61,916 to ₹95,700. Among mainstream chipsets, Snapdragon powers the largest number of devices (407 phones) while maintaining a strong balance of performance (2.59 GHz) and customer satisfaction (81.62 rating). In contrast, budget-focused processors such as Unisoc (1.65 GHz) and Spreadtrum (1.35 GHz) are associated with lower-priced devices (₹8,160 and ₹5,569 respectively) and lower average ratings. The results indicate a clear relationship between processor performance, smartphone pricing, and perceived product quality.

Q23. RAM Tier Distribution

-- Group by RAM capacity; bucket anything outside common tiers.

Insight: 8 GB RAM is the most common configuration in the dataset, accounting for 339 smartphones, followed by 6 GB (234 devices) and 4 GB (217 devices). As RAM capacity increases, both smartphone prices and storage capacities rise significantly. Devices with 4 GB RAM average ₹16,124 and 86 GB storage, while 12 GB RAM models average ₹72,911 and 292 GB storage. The premium segment is dominated by 12 GB and 16 GB RAM devices, which offer the highest storage capacities (292–341 GB) and command substantially higher prices. These findings indicate that manufacturers position higher RAM configurations as premium offerings, typically bundled with larger storage capacities and flagship-level features.

Q24. Fully-Loaded Phones — Multi-Feature Filter

-- 5G + NFC + IR Blaster + Fast Charging + 120Hz+ + 8GB+ RAM

-- Also shows % of total catalogue.

Insight: Only 4.18% of the smartphone catalogue meets all premium feature criteria simultaneously: 5G, NFC, IR Blaster, Fast Charging, 120Hz+ display, and at least 8 GB RAM. The segment is heavily dominated by Xiaomi, with models such as the Xiaomi Mix Fold 2 5G (₹106,990), Xiaomi 13 Ultra 5G (₹71,999), and Xiaomi Redmi Note 12 Explorer (₹24,999, Rating: 89). Other notable brands include Vivo, iQOO, Poco, and Honor. The results suggest that combining all flagship-level features into a single device remains relatively rare, with only a small subset of smartphones positioned as truly feature-complete offerings.

Q25. High-Resolution Rear Camera Phones (100+ MP)

Insight: The dataset contains a significant number of smartphones featuring 100+ MP rear cameras, highlighting the industry's strong focus on camera innovation. The highest camera resolution observed is 200 MP, offered by brands such as Samsung, Xiaomi, Motorola, Nokia, Honor, and Infinix.

Notably, 200 MP cameras are available across a wide price range, from the Xiaomi Redmi Note 12 Explorer (₹24,999) to premium flagship devices like the Samsung Galaxy S24 Ultra (₹119,990).

Xiaomi dominates this category with the largest number of 108 MP and 200 MP devices, demonstrating its aggressive camera-centric product strategy. These findings suggest that ultra-high-resolution photography is no longer exclusive to flagship smartphones and has increasingly expanded into the upper mid-range segment.

Note: Ultra-high-resolution cameras have become a key competitive feature across both premium and mid-range smartphones, making advanced mobile photography accessible to a broader consumer base.

LEVEL 4 – ADVANCED SQL ANALYSIS

Q26. Highest-Rated Smartphone Per Brand

-- Uses ROW_NUMBER() partitioned by brand.

-- If two phones tie for top rating, both appear (use RANK).

Insight: Using a window function (RANK()), the analysis identified the highest-rated smartphone(s) for each brand, including ties. Several brands achieved an outstanding rating of 89, including Google (Pixel 6 Pro), OnePlus (OnePlus 11 5G), Samsung (Galaxy S23 Plus, Galaxy Z Fold 3), Xiaomi (Xiaomi 14, Redmi Note 12 Explorer), Motorola (ThinkPhone, Edge 40 Pro 5G), Huawei (Mate Xs 2), and Sony (Xperia 1 II). The results show that top-rated smartphones are distributed across multiple brands and price segments, ranging from the Xiaomi Redmi Note 12 Explorer (₹24,999) to premium foldables such as the Samsung Galaxy Z Fold 3 (₹110,999). This indicates that exceptional customer satisfaction is not exclusive to ultra-premium devices.

Business Takeaway

High customer satisfaction is achievable across different price segments and brands. Premium pricing may provide advanced features, but it does not guarantee a higher rating than well-positioned mid-range devices.

Q27. Price Ranking Within Each Brand

-- RANK() with PARTITION BY brand; ties get the same rank.

Insight:

Price rankings reveal significant pricing variation within smartphone brands.

- Apple's most expensive model is the iPhone 14 Pro Max (1TB) at ₹182,999, while the lowest-priced model is the iPhone 9 at ₹29,990 — a price gap of over ₹153,000.
- Samsung shows one of the widest product portfolios, ranging from the Galaxy Z Fold 4 (₹163,980) to the Galaxy A01 Core (₹4,999), serving both premium and budget segments.
- Xiaomi's highest-priced model is the Redmi K20 Pro Signature Edition at ₹480,000, far above its mainstream lineup, making it a major outlier in the dataset.

- Premium-focused brands such as Apple, OnePlus, Google, and ASUS have relatively smaller price ranges compared to diversified brands like Samsung, Xiaomi, Vivo, and Oppo.
- The ranking analysis highlights how different brands position products across flagship, mid-range, and budget categories to target multiple customer segments.

Q28. Price Percentile Within Brand — Top Tier Flag

-- PERCENT_RANK() = 0 for cheapest, 1 for most expensive per brand.

-- Top 10% (PERCENT_RANK >= 0.90) flagged as 'Top Tier'.

Insights:

The PERCENT_RANK() function successfully identified the top 10% most expensive smartphones within each brand as **Top Tier** products. Apple's premium portfolio is led by devices such as the **iPhone 14 Pro Max (1TB) priced at ₹182,999** and **iPhone 13 Pro Max (1TB) priced at ₹179,900**. Brands including **Samsung, OnePlus, Vivo, Oppo, iQOO, and Motorola** show clearly defined flagship segments concentrated in the highest price percentiles. The analysis demonstrates how smartphone manufacturers use premium models to create distinct product tiers and target high-value customer segments.

Q29. Phones Priced Above Their Own Brand's Average

-- Classic correlated subquery pattern.

-- Inner query references outer query's brand_name.

Insight:

- The analysis identified smartphones priced above their respective brand's average selling price, highlighting each brand's premium product segment.
- Apple shows a strong premium positioning, with flagship models such as the iPhone 14 Pro Max (1TB) priced at ₹182,999, significantly above the brand average price of ₹95,966.
- Samsung, Vivo, Xiaomi, OnePlus, and Oppo have extensive premium portfolios, with numerous devices priced well above their brand averages, indicating a broad presence across flagship and ultra-premium segments.
- Xiaomi exhibits one of the largest pricing gaps, with the Redmi K20 Pro Signature Edition priced at ₹480,000 compared to the brand average of ₹27,961, reflecting the presence of ultra-premium outlier products.
- The results reveal that most leading smartphone brands follow a tiered pricing strategy, maintaining premium flagship models alongside mainstream offerings.

Q30. Top 3 Rated Phones Per Brand — Handling Ties

-- DENSE_RANK() does not skip numbers after a tie.

-- e.g. 89, 89, 86 → ranks 1, 1, 2 (not 1, 1, 3)

Insight:

- Using DENSE_RANK(), the analysis identified the top three rating tiers within each smartphone brand, while correctly handling rating ties.
- Several brands, including Samsung, Xiaomi, OnePlus, Motorola, Google, Huawei, and Sony, have smartphones achieving the highest observed rating of 89, indicating strong customer satisfaction across multiple manufacturers.
- Due to tied ratings, some brands have more than three phones in the result set. For example, Samsung has multiple devices ranked #1 with a rating of 89, including the Galaxy S23 Plus and Galaxy Z Fold 3.
- Xiaomi's top-rated portfolio includes Xiaomi 14, Xiaomi 13 5G, and Redmi Note 12 Explorer, all rated 89, showing strong performance across both flagship and value-premium segments.
- The results highlight that highly rated smartphones are distributed across different brands and price ranges, suggesting that customer satisfaction depends on overall value and user experience rather than price alone.

Q31. LAG and LEAD — Price Gap Between Adjacent Phones

- All 980 phones ordered by price ascending.
- Shows each phone's price, the one below, the one above,
- and the gap to the next phone up.

Insights:

- The smartphone market is highly concentrated in the ₹10,000–₹30,000 range, where many adjacent models have price gaps of ₹0–₹500, indicating intense competition and aggressive pricing.
- Multiple phones share identical price points (₹9,999, ₹14,999, ₹19,999, ₹29,999, etc.), showing that brands deliberately cluster around popular psychological price bands.
- At the entry-level segment (< ₹10,000), price differences are very small, often below ₹200, suggesting limited product differentiation and price-sensitive customers.
- The premium segment (> ₹1 lakh) shows significantly larger price jumps. For example, the gap rises from ₹239,999 to ₹480,000 (+₹240,001) and from ₹480,000 to ₹650,000 (+₹170,000), indicating luxury and niche flagship categories.
- Vertu Signature Touch (₹650,000) stands as an extreme outlier with no higher-priced competitor in the dataset.
- The analysis reveals clear market tiers: Budget (<₹15K), Mid-Range (₹15K–₹30K), Premium (₹30K–₹1L), and Ultra-Luxury (>₹1L), with price gaps widening substantially as products move up the ladder.

Q32. Brand Consistency Score — Coefficient of Variation

- $CV = \text{STDDEV_POP}(\text{price}) / \text{AVG}(\text{price}) * 100$
- Low CV = brand prices all models similarly (consistent)
- High CV = brand spans many segments (diverse portfolio)
- Only brands with 5+ phones included.

Insight:

- itel is the most price-consistent brand in the dataset with a CV of just 14.55%, indicating that its models are concentrated within a narrow budget price range.
- Nothing (18.55%), Nubia (18.33%), and ASUS (26.16%) also maintain relatively focused pricing strategies, targeting specific customer segments rather than the entire market.
- Apple shows moderate pricing consistency with a CV of 44.72%, suggesting a premium-focused portfolio despite having multiple product tiers.
- Samsung (89.48%), Huawei (91.43%), and especially Xiaomi (162.89%) exhibit highly diversified pricing structures, spanning budget, mid-range, premium, and ultra-premium segments.
- Xiaomi has the highest variation in the dataset, driven by products ranging from affordable Redmi devices to ultra-premium models such as the Redmi K20 Pro Signature Edition (₹480,000).
- The analysis reveals two distinct strategies in the smartphone market: focused brands that serve a specific segment and diversified brands that compete across multiple price categories.

Q33. Brand Ranking Within Price Segments (CTE + Window)

- CTE: segment each phone and compute avg rating per brand-segment
- Main SELECT: RANK() brands within each segment

Insights:

- The analysis ranked brands based on average ratings within each price segment, enabling a fair comparison among competitors targeting similar customer groups.

- In the Budget segment, BLU (85) and Doogee (84) achieved the highest average ratings, while larger brands such as Xiaomi (73) and Samsung (71.3) ranked lower despite having significantly more models.
- In the Mid-Range segment, ASUS (87) and Nothing (85.2) led customer satisfaction rankings, outperforming larger competitors such as Xiaomi (82.62) and Samsung (80.43).
- In the Premium segment, Tecno and ZTE ranked first with an average rating of 89, while Motorola (88.5) and Xiaomi (88) also performed strongly.
- In the Flagship segment, Sony and Leitz secured the top position with an average rating of 89, whereas Apple ranked 9th with an average rating of 79.44, despite being one of the most premium-priced brands.
- The results indicate that higher pricing does not automatically lead to higher customer ratings, and several smaller brands outperform major manufacturers within their respective price categories.

Q34. Composite Smartphone Score — Leaderboard Top 20

-- Formula (max values from dataset):

- $\text{rating} / 89 \times 40$ (weight: 40%)
- $\text{battery_capacity} / 22000 \times 20$ (weight: 20%)
- $\text{ram_capacity} / 18 \times 20$ (weight: 20%)
- $\text{primary_camera_rear} / 200 \times 20$ (weight: 20%)
- Excludes any phone with NULL in these four fields.

Insights:

- Doogee V Max achieved the highest composite score (83.68), driven by its massive 22,000 mAh battery, 12GB RAM, 108MP camera, and 88 rating, demonstrating the impact of balancing multiple high-performance features.
- Xiaomi dominated the leaderboard with multiple entries, including the Redmi Note 12 Pro Plus (76.96), 12T Pro 5G (73.43), and Redmi Note 12 Explorer (72.80), indicating a strong combination of camera quality, battery life, and customer satisfaction.
- Despite its premium price of ₹119,990, the Samsung Galaxy S24 Ultra ranked 3rd with a score of 76.17, showing that higher pricing alone does not guarantee the top composite ranking.
- Gaming-focused devices such as the ASUS ROG Phone 7 (70.96), ASUS ROG Phone 6 Pro (70.01), and Lenovo Legion Pro 2 (68.72) performed strongly due to their high RAM capacities and strong ratings.
- Several mid-priced smartphones, including the Infinix Zero Ultra (72.98) and Oukitel WP19 (72.13), outperformed many flagship devices, suggesting that feature-rich phones can deliver exceptional value without ultra-premium pricing.
- The composite score approach provides a holistic performance benchmark by combining rating (40%), battery (20%), RAM (20%), and camera quality (20%) into a single comparable metric.

Q35. Multi-CTE Tier Dominance Analysis

-- Which brand has the MOST models in each price segment?

-- CTE 1 → segment every phone

-- CTE 2 → count models per brand per segment

-- CTE 3 → rank brands within each segment

-- Final → show only the dominant brand (rank = 1) per segment

Insights:

- **Xiaomi dominates the Budget segment with 56 models**, demonstrating a strong focus on affordable smartphones and broad market coverage.
- **Xiaomi also leads the Mid-Range segment with 61 models**, making it the most diversified brand in the largest and most competitive price category.

- In the **Premium segment**, **Samsung ranks first with 20 models**, reflecting its strong presence in the ₹40,000–₹80,000 range through its Galaxy S, A, and Fold series lineup.
- **Apple dominates the Flagship segment with 25 models**, highlighting its strategy of concentrating heavily on the ultra-premium smartphone market.
- The analysis reveals distinct market-positioning strategies: **Xiaomi focuses on scale and volume in Budget and Mid-Range categories, Samsung leads the Premium segment, while Apple dominates the Flagship space.**
- Overall, the smartphone market shows that different brands use different portfolio strategies rather than competing equally across all price segments.

Project Summary

This project analyzed a dataset of approximately 980 smartphone models from multiple global brands to uncover pricing patterns, product strategies, customer satisfaction trends, and technology adoption across the smartphone market.

The analysis began with market-level exploration of pricing, ratings, storage, RAM, battery capacity, camera specifications, and operating systems. It then expanded into brand-level benchmarking, identifying premium brands, value-for-money brands, highest-rated brands, and brands with the most diversified product portfolios.

Feature-focused analyses revealed important market trends such as 5G adoption rates, processor market share, fast-charging capabilities, refresh-rate preferences, expandable storage strategies, and high-resolution camera adoption. The project also examined relationships between smartphone specifications and pricing, helping identify the features that contribute most strongly to premium positioning.

Advanced SQL techniques were used to perform ranking, segmentation, percentile analysis, brand benchmarking, portfolio dominance analysis, and composite scoring. Window functions such as RANK(), DENSE_RANK(), PERCENT_RANK(), LAG(), and LEAD() enabled deeper analytical insights beyond traditional aggregation queries.

Key findings showed that:

- Xiaomi dominates the Budget and Mid-Range smartphone segments.
- Samsung maintains a strong Premium segment presence.
- Apple dominates the Flagship segment.
- Several smaller brands achieve customer ratings equal to or higher than premium manufacturers.
- Higher pricing does not always lead to higher customer satisfaction.
- 5G adoption is becoming a key differentiator among leading brands.
- Smartphone brands follow distinct portfolio strategies depending on their target market.

Overall, the project demonstrates how SQL can be used not only for data retrieval but also for advanced business analysis and decision-making. By combining descriptive analytics, segmentation, ranking methods, statistical measures, and business interpretation, the project provides a comprehensive view of the smartphone market and showcases practical Data Analyst skills that can be applied in real-world business environments.
